ENTERPRISE MLOPS ENGINEERING

End-to-End Content Catalog Recommendation Pipeline

Automated metadata vectorization, similarity search, and continuous orchestration on Google Cloud Platform.

ENGINEER
Ruby Ineba Briggs

PROJECT ID
`netflixtitles-512209`

REGION
`us-central1`

EXECUTIVE SUMMARY

Operational Impact & Target Milestones

8,807
Catalog Records Staged

768
Vector Embedding Dimensions

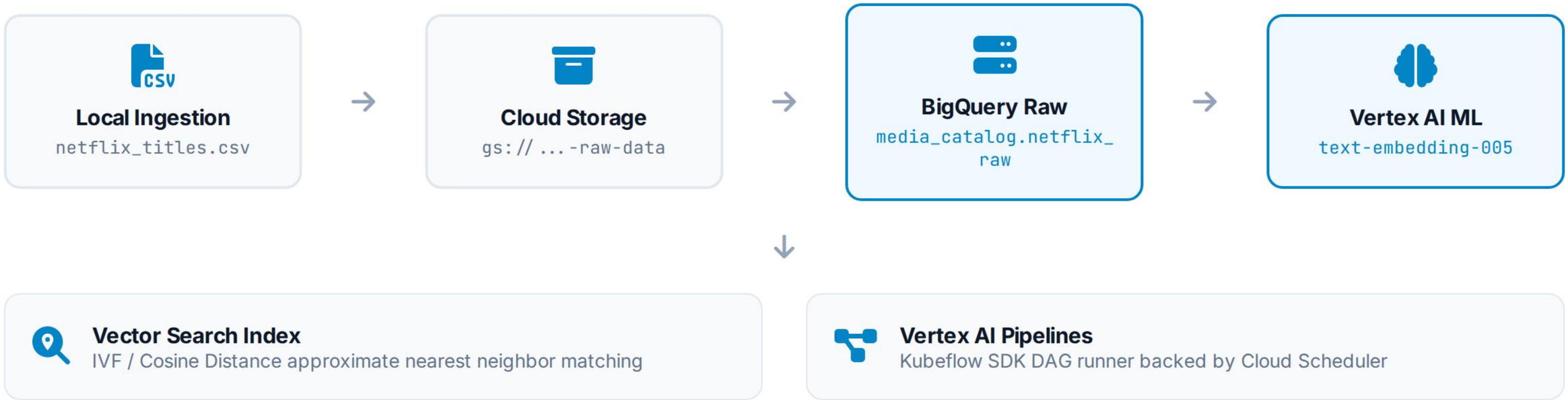
53s
Pipeline Execution Time

Active
Weekly Cron Recurrence

Successfully converted raw local CSV assets into a fully automated, cloud-native vector search and recommendation framework on Google Cloud Platform.

ARCHITECTURE DESIGN

End-to-End MLOps Data Flow



EXECUTION BREAKDOWN

Four-Phase MLOps Lifecycle

01

Data Ingestion

Staged raw CSV to Cloud Storage and loaded structured schema into BigQuery.

02

Feature Embeddings

Generated 768-dim dense vectors using Vertex AI text-embedding-005.

03

Vector Search

Built IVF Cosine index enabling real-time semantic catalog recommendations.

04

Automated DAG

Deployed Kubeflow Pipelines runner with weekly Cloud Scheduler triggers.

VALIDATION & VERIFICATION

Pipeline Job Execution & Monitoring

```
# Pipeline Job Verification Output

Pipeline Name: netflix-recommendation-pipeline
Run ID: 20261020013915
State: PIPELINE_STATE_SUCCEEDED
Container Image: python:3.11
Execution Duration: 53 seconds
Schedule ID: 3489148418142502912

✓ 1/1 DAG steps completed successfully.
```

Zero Schema Errors
Parsed multiline text fields safely using explicit quote handling.

Deterministic Runtime
Pinned explicit base image python:3.11 for KFP SDK stability.

Automated Cron Trigger
Pipeline scheduled for weekly runs every Sunday at midnight.



MLOps Pipeline Live

The catalog embedding feature store and similarity recommendation engine are fully deployed and operational.

Status: Active & Scheduled